

14 Singular Value Decomposition



14.3 Four Flavors of SVD

full, economy, compact, and truncated, seen through block-matrix heatmaps

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Full SVD: both $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices; $\mathbf{S}$ matches the shape of the original matrix

2

Economy SVD: $\mathbf{S}$ becomes a diagonal square matrix, and $\mathbf{U}$ becomes a semi-orthogonal matrix

3

Compact SVD: only when a tall matrix is not full column rank can the economy form be compressed further, dropping the zero singular values

4

Truncated SVD: keeping the top p components to approximate the original matrix — a common data-compression technique

From full to truncated, four SVD forms trimmed step by step

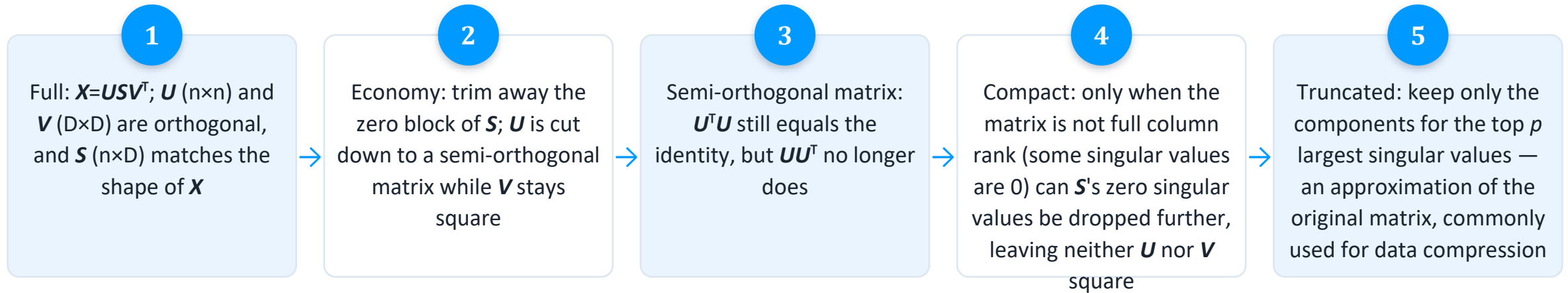


Figure 1 · Full singular value decomposition

- $X = USV^T$ U ($n \times n$) and V ($D \times D$) are both orthogonal matrices; S ($n \times D$) matches the shape of X

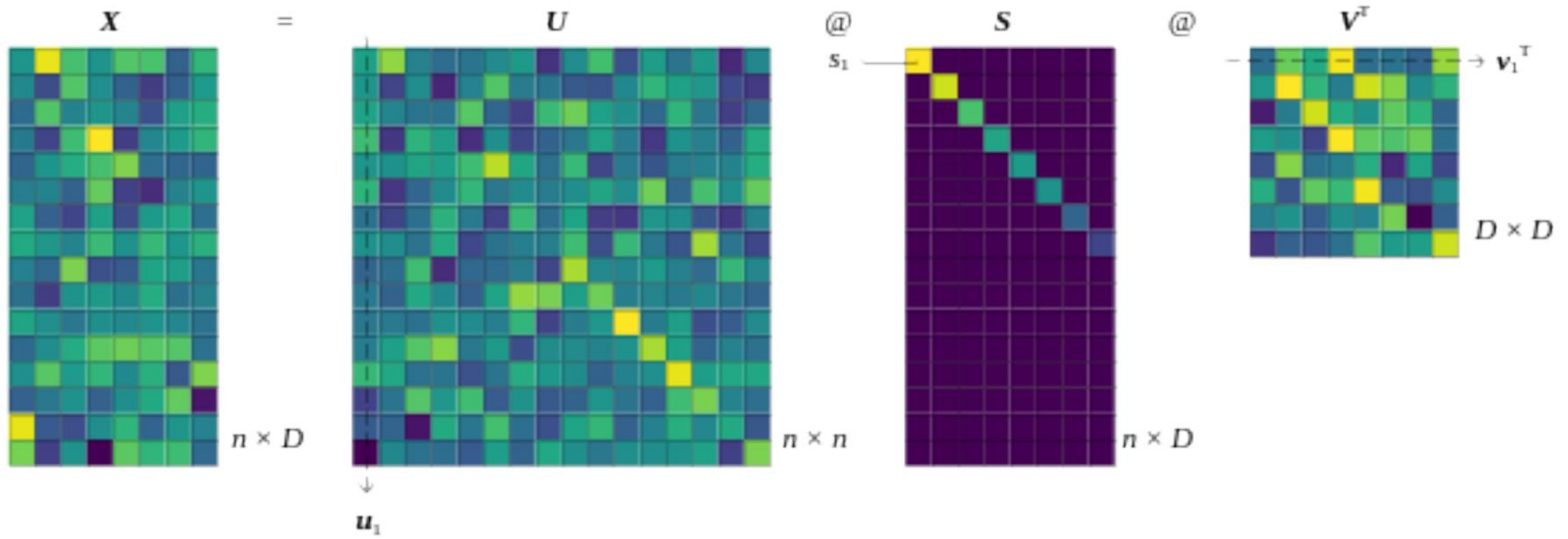


Figure 2 · SVD, from full to economy form, X with full column rank

- S drops its zero block and becomes a diagonal square matrix; U is trimmed to the semi-orthogonal $U_{n \times D}$, while V stays square

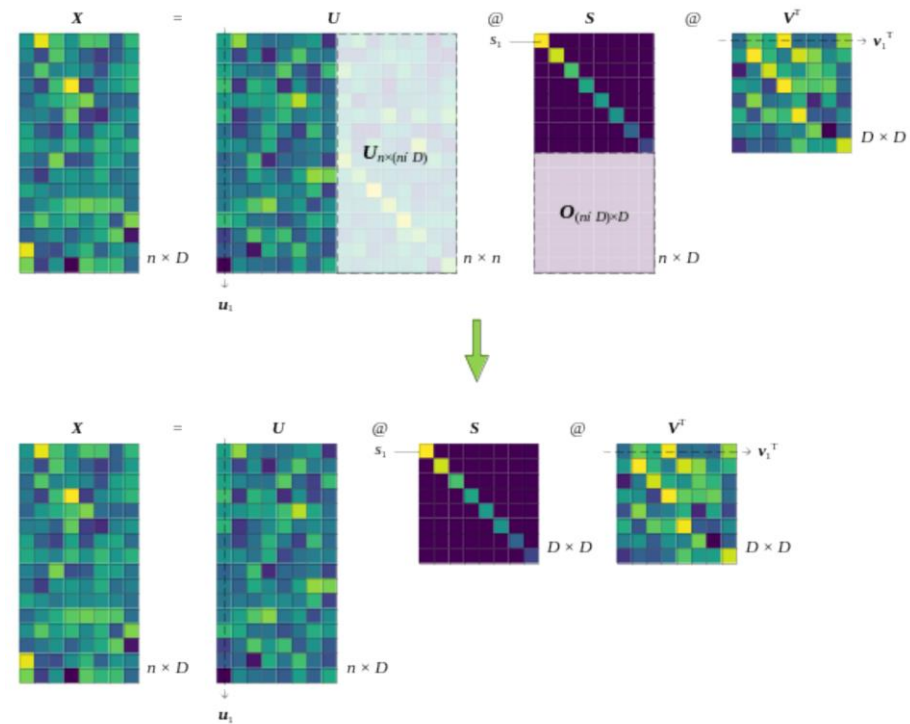


图 2. SVD, 从完全型到经济型, X 列满秩

Figure 3 · The column spaces of $U_{n \times D}$ and $U_{n \times (n-D)}$ are orthogonal complements

- Cut the orthogonal matrix $U_{n \times n}$ at column D — the two resulting column spaces are orthogonal complements of each other

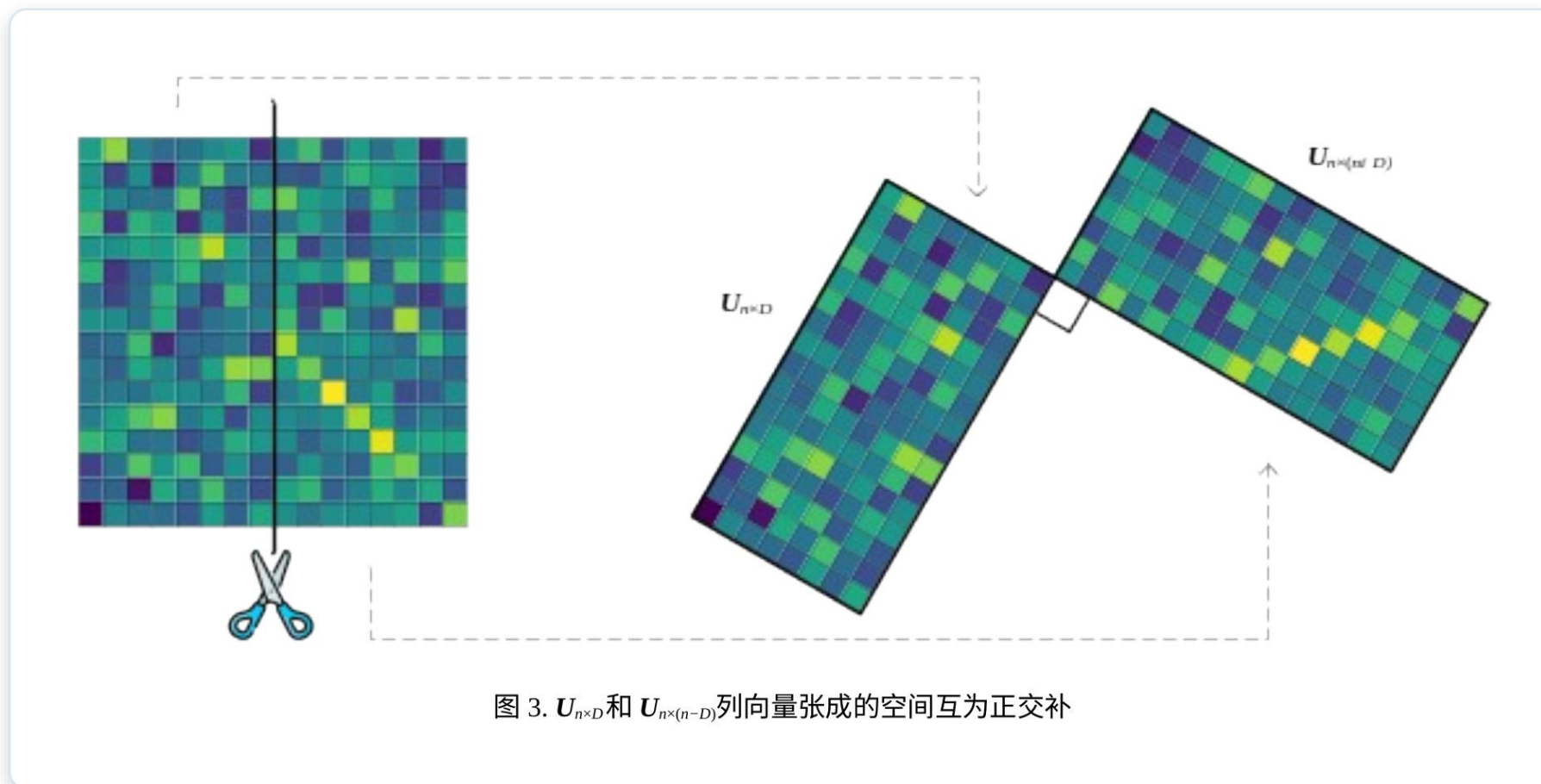


Figure 4 · $U_{n \times D}^T$ times $U_{n \times D}$ is still the identity matrix

- $U_{n \times D}$ is not square, but its columns are still unit-orthogonal, so $U_{n \times D}^T U_{n \times D} = I_{D \times D}$

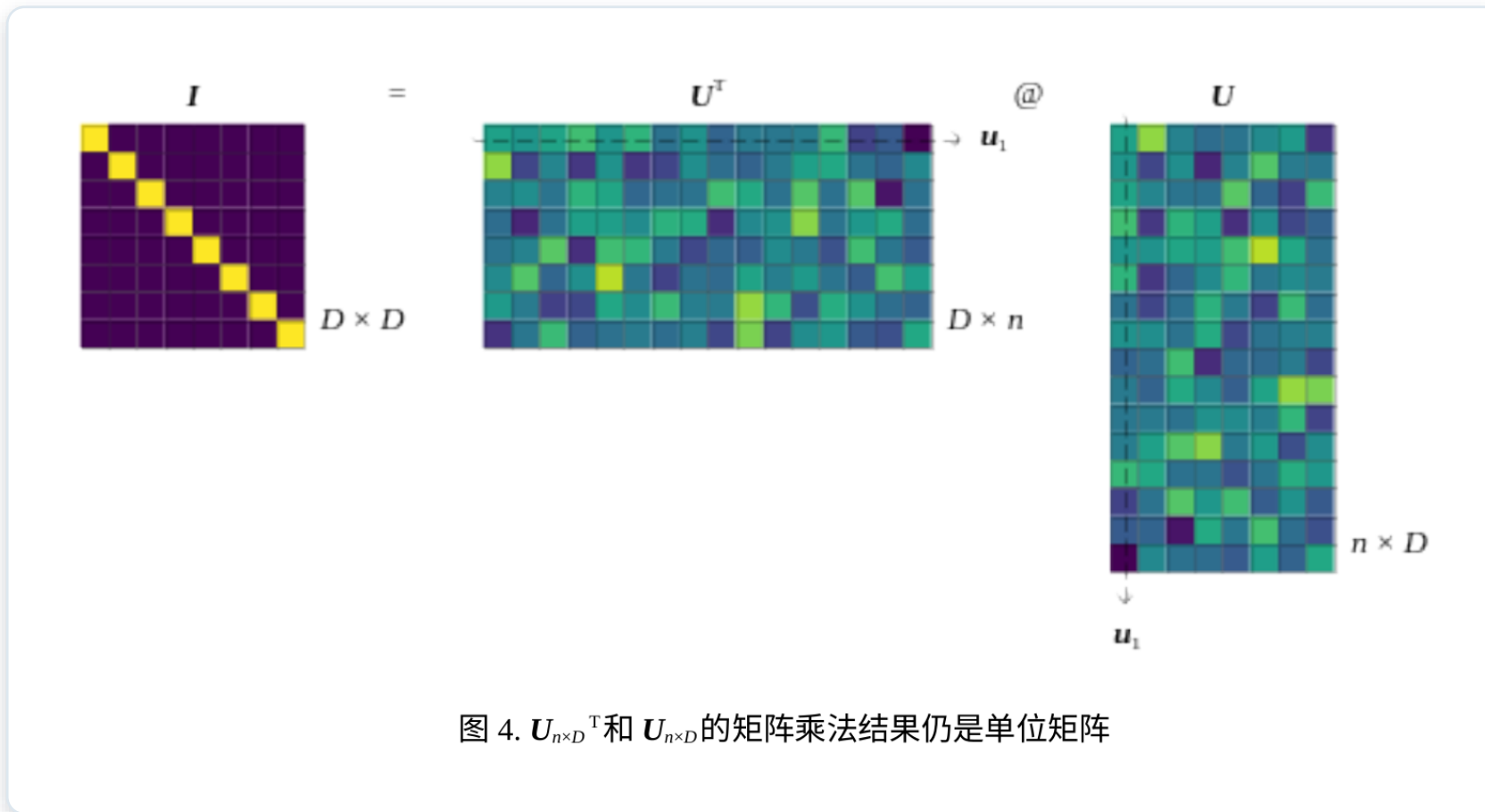


Figure 5 · $U_{n \times D}$ times $U_{n \times D}^T$ is no longer the identity matrix

- $U_{n \times D} U_{n \times D}^T = I_n - (u_{D+1} u_{D+1}^T + \dots + u_n u_n^T)$; work this out from the perspective of an orthogonal projection matrix

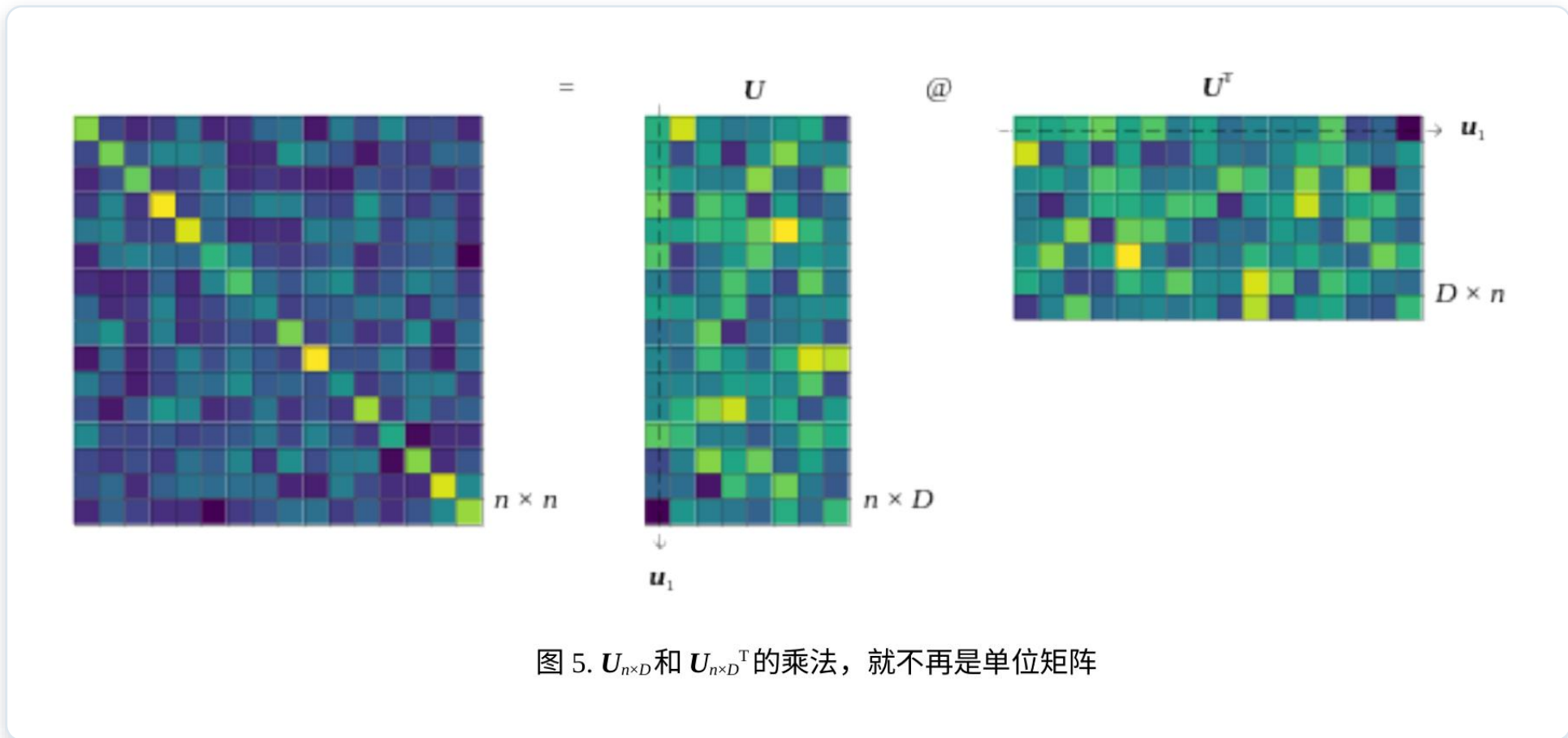


Figure 6 · Full SVD, for a wide matrix

- The full SVD of a wide matrix takes a different form than that of a tall matrix — work it out for yourself

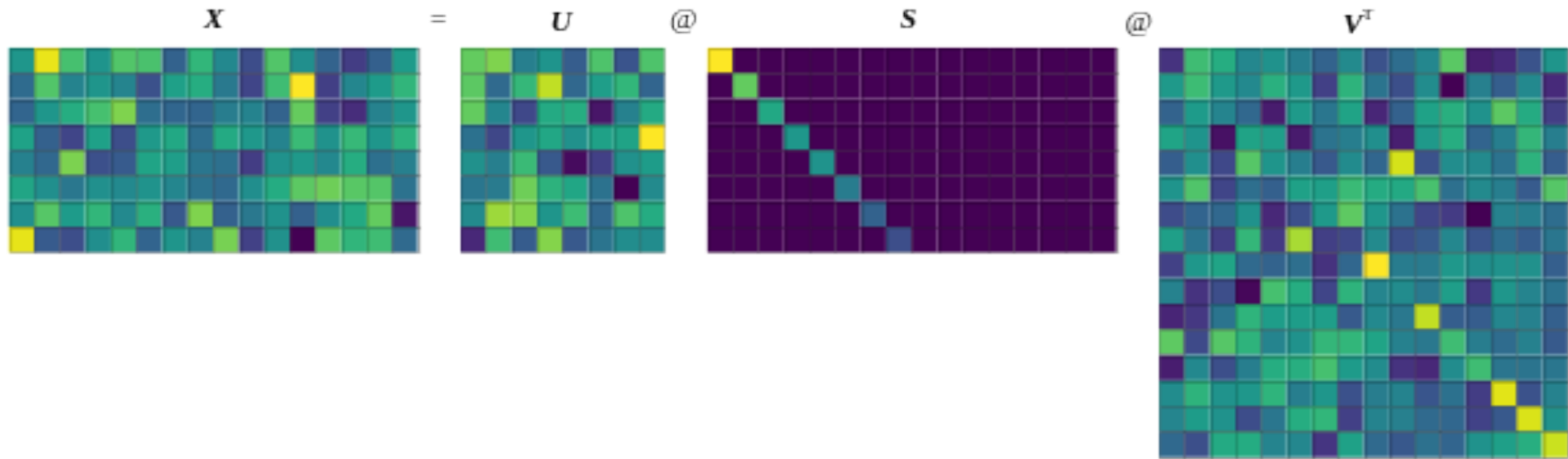


图 6. 完全型 SVD 分解，扁平矩阵

Figure 7 · SVD, from full to economy form, for a wide matrix

- Compare Figure 2 and Figure 7: matrix V always stays square — the one part unchanged between full and economy SVD

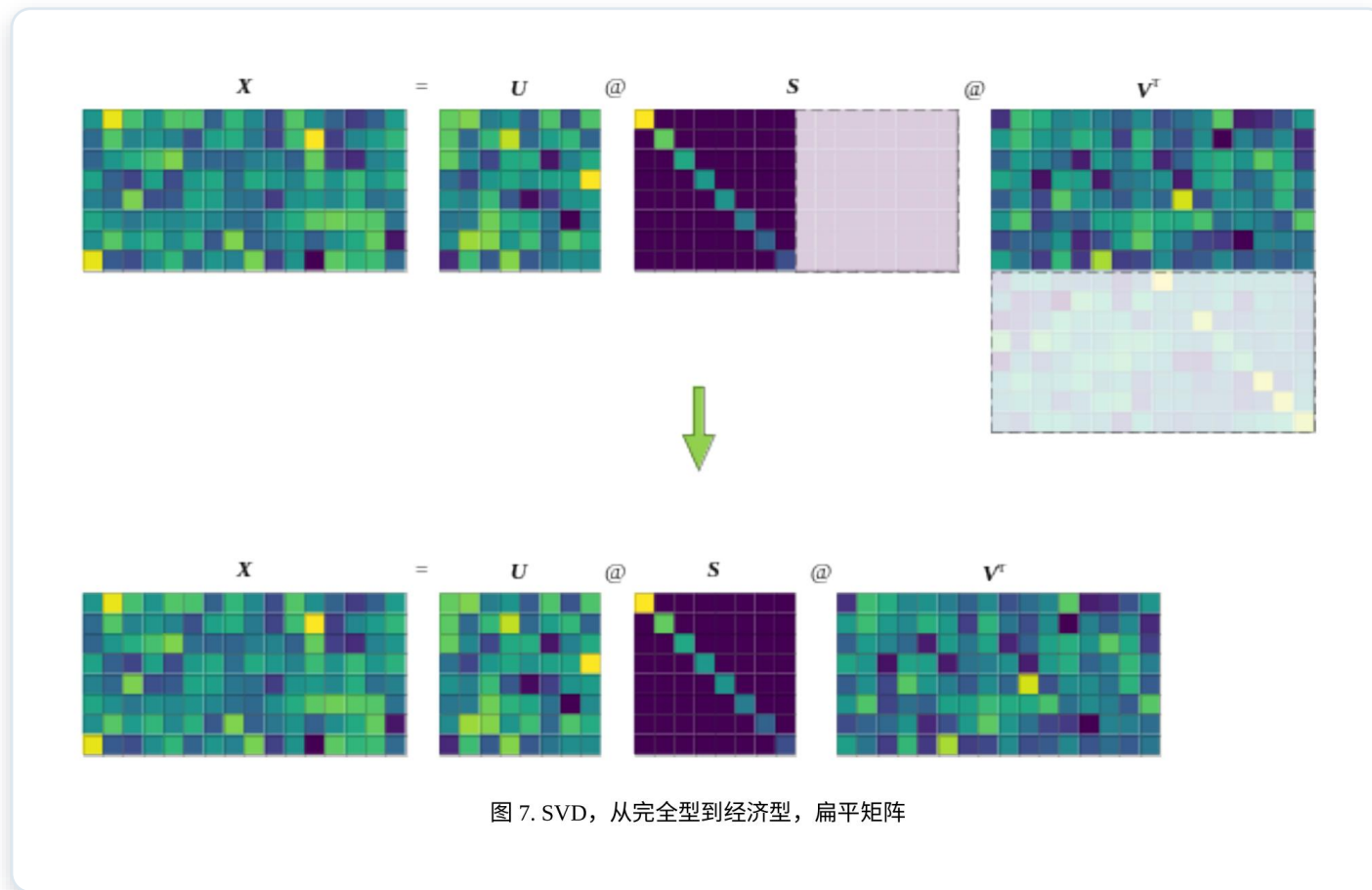


Figure 8 · SVD, from economy to compact form, X not full column rank

- Only when $\text{rank}(X)=r<D$ can S be split into 4 blocks; dropping the all-zero parts gives $X=U_{n \times r} S_{r \times r} (V_{D \times r})^T$, and neither U nor V is square any more

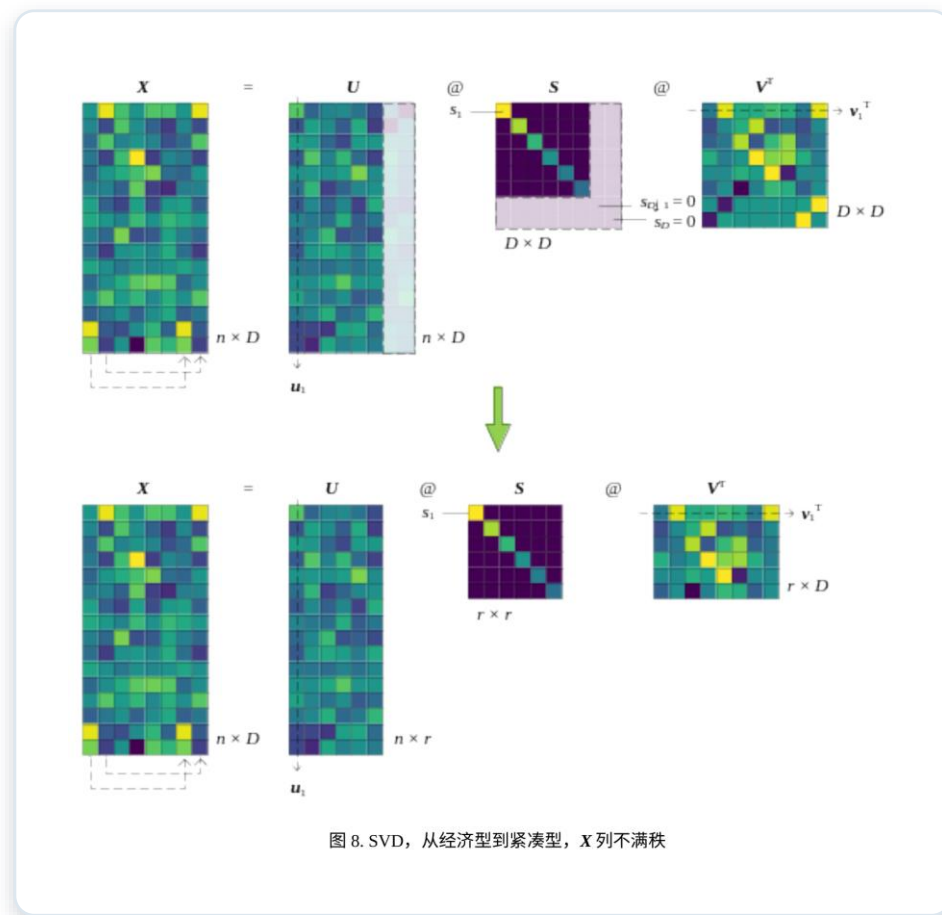


图 8. SVD, 从经济型到紧凑型, X 列不满秩

The number of nonzero singular values sets the rank of the matrix

① Number of nonzero singular values = $\text{rank}(\mathbf{X})$ = column rank = row rank

② Column rank: the largest number of linearly independent column vectors in the matrix

③ Row rank: the largest number of linearly independent row vectors in the matrix

④ The column space of $\mathbf{U}$ equals the column space of $\mathbf{X}$; the row space of $\mathbf{V}^T$ equals the row space of $\mathbf{X}$

Figure 9 · Approximating the original data with truncated SVD

- Keeping the top p singular values: $\mathbf{X} \approx \hat{\mathbf{X}} = \mathbf{U}_{n \times p} \mathbf{S}_{p \times p} (\mathbf{V}^T_{p \times D})^T$ — an approximation, not an exact reconstruction

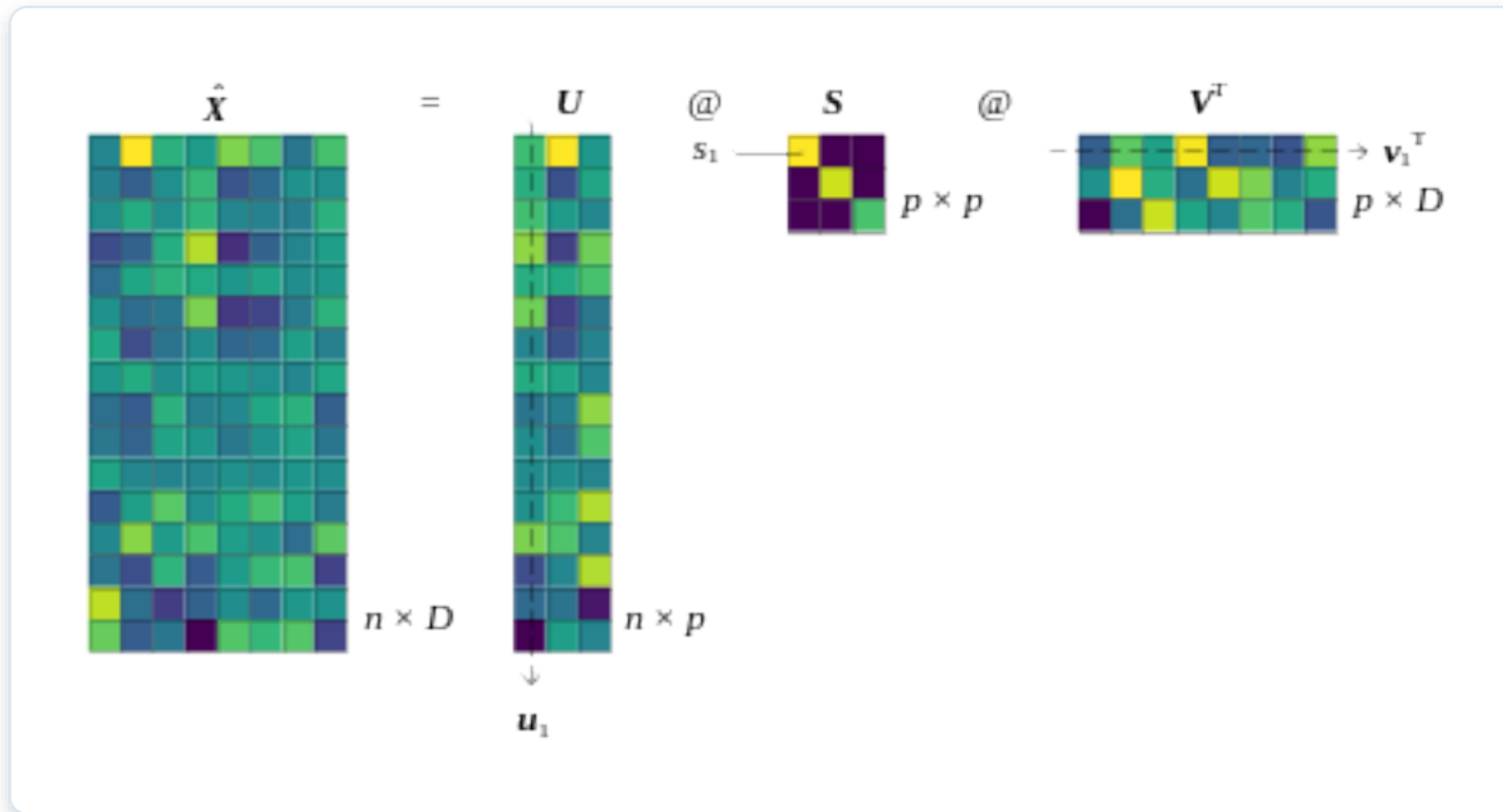


Figure 10 · The error

- $X_{\text{error}} = X - \hat{X}$ — where does this error come from? That's the topic of the next section

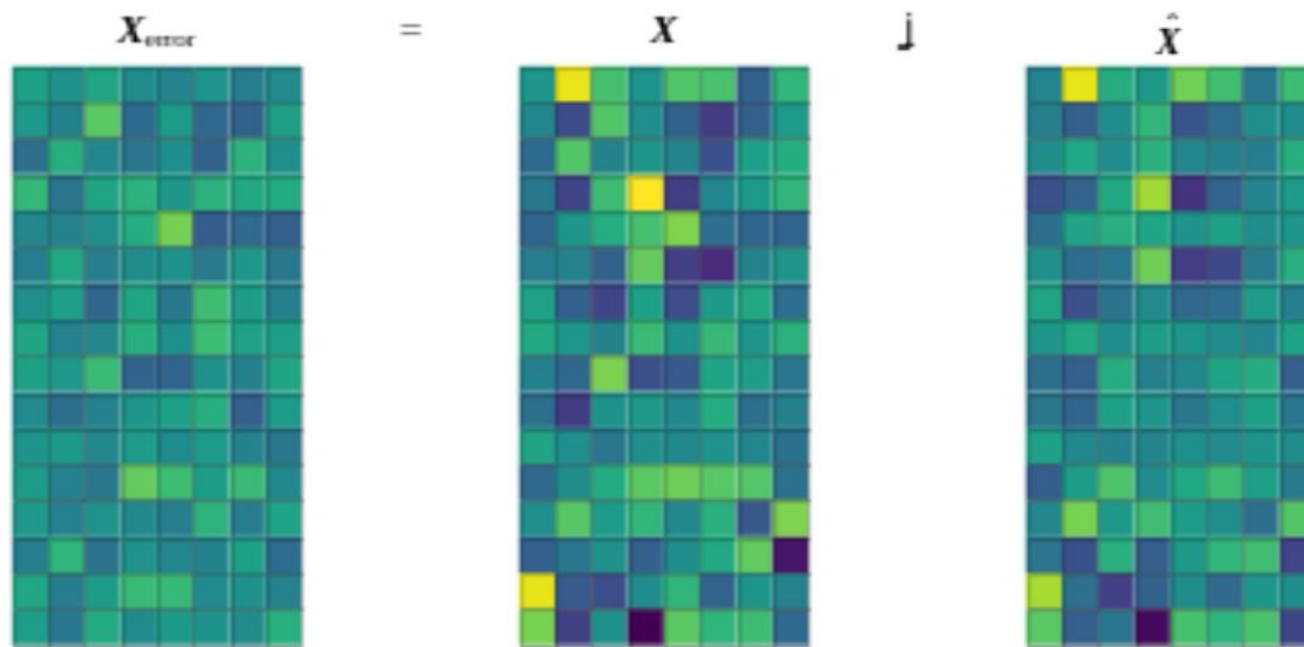


图 10. 误差

SUMMARY

Four flavors of SVD, trimmed step by step down to a data's essential structure

- ① Full: both $\mathbf{U}$ and $\mathbf{V}$ are square and orthogonal; $\mathbf{S}$ matches the shape of $\mathbf{X}$
- ② Economy: $\mathbf{S}$ drops its zero block to become a diagonal square matrix; $\mathbf{U}$ is trimmed to semi-orthogonal
- ③ Compact: exists only when not full column rank, further dropping the zero singular values in $\mathbf{S}$
- ④ Truncated: keeps only the top p largest singular values — an approximation ($\approx$) of the original matrix, used for compression
- ⑤ Number of nonzero singular values = rank of the matrix = column rank = row rank
- ⑥ $\mathbf{V}$ for a tall matrix, or $\mathbf{U}$ for a wide matrix, always stays square in economy SVD

Key Terms in This Section

Full SVD $X=USV^T$, with U and V both orthogonal square matrices, and S matching the shape of X

Compact SVD Only when the matrix is not full column rank, further dropping the zero singular values in S

Semi-orthogonal Matrix Its columns are unit-orthogonal but it has fewer columns than rows; $U^T U = I$, but $U U^T \neq I$

Economy SVD S drops its extra zero block to become a diagonal square matrix; U is trimmed to semi-orthogonal accordingly

Truncated SVD Keeps only the components for the top p largest singular values — an approximation ($\approx$) of the original matrix

Rank of a Matrix The number of nonzero singular values, equal to both the column rank and the row rank

FURTHER READING

Further Reading

Review · Chapter 3, Section 5

Block matrix multiplication — the foundation for this section's derivations

Review · Chapter 9, First Three Sections

Orthogonal matrices and orthogonal projection matrices

Review · Section 14.2, A Geometric View of SVD

Understanding rotation and scaling from a geometric perspective

Open-source Code Repository

Code and figures for the entire book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1 Work out the full SVD of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$, then rewrite the result as its economy SVD.

Q3 Compute the row rank and column rank of matrix A in both Q1 and Q2.

Q5 Compute the Gram matrix AA^T for the A in Q1 and Q2, along with its row rank and column rank.

Q7 In Python, verify that in the economy SVD of the Q1 matrix A , $U_{n \times D}$ satisfies $U^T U = I$ but $U U^T \neq I$.

Q9 Write a Python function that automatically decides whether a matrix admits a compact SVD (i.e., whether it is full column rank).

Q2 Write out the full SVD, economy SVD, and compact SVD of $A = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$.

Q4 Compute the Gram matrix $A^T A$ for the A in Q1 and Q2, along with its row rank and column rank.

Q6 In Python, compute `np.linalg.svd(A, full_matrices=True/False)` for the matrix A in Q1, and compare the shapes of U , S , V .

Q8 In Python, verify that one singular value of the Q2 matrix A equals 0, and use that to write out its compact SVD.

Exercises Q10 – Q17

Q10 Design your own 4×2 tall matrix (full column rank), work out its full and economy SVD, and verify in Python.

Q12 In Python, verify that for any matrix X , the number of nonzero singular values equals $\text{rank}(X)$, the column rank, and the row rank.

Q14 Vary p in Q13 ($p=1,2,3,4,5$), plot the error's Frobenius norm against p , and observe the trend.

Q16 Explain why truncated SVD's " $\approx$ " cannot be written as " $=$ ", illustrating with a concrete numerical example.

Q11 Design your own 2×4 wide matrix, work out its full and economy SVD, and compare it with the tall-matrix case.

Q13 In Python, generate a random 5×5 matrix, compute its truncated SVD with $p=3$, and compute the Frobenius norm of the error matrix X_{error} .

Q15 Use Matplotlib to plot the heatmaps of X , $\hat{X}$ (at $p=3$), and X_{error} from Q13, in the style of Figure 10.

Q17 In Python, verify that V always stays square in economy SVD, regardless of whether the original matrix is tall or wide.

Exercises Q18 – Q25

Q18 In Python, load a grayscale image, compress it with truncated SVD at $p=5$, 20, and 50, and compare the results.

Q20 In Python, verify that the eigenvalues of $A^T A$ equal the squares of the singular values of A .

Q22 Drawing on orthogonal projection matrices from Chapter 9, explain why $U \times D \times U^T$ in Figure 5 equals a projection matrix.

Q24 Design a 3×3 matrix with exactly two nonzero singular values, write out its compact SVD, and verify it in Python.

Q19 Explain why truncated SVD is a common data-compression technique, illustrating with the image example from Q18.

Q21 In Python, verify that AA^T and $A^T A$ share exactly the same nonzero eigenvalues.

Q23 In Python, compute the compact SVD of a full-column-rank matrix and a rank-deficient one, and compare the shapes of U , S , V .

Q25 Summarize how the four SVD forms differ in storage (the element counts of U , S , V), and explain why truncated SVD is the most economical.

10 Prompts to Explore with AI

- 1 Explain the differences and relationships among full, economy, compact, and truncated singular value decomposition.
- 2 Explain why the U matrix in economy SVD is called a “semi-orthogonal matrix”, and how it differs from an orthogonal matrix.
- 3 Explain why $U^T U$ equals the identity matrix, but $U U^T$ does not.
- 4 Explain why compact SVD only exists when a matrix is not full column rank.
- 5 Explain why the number of nonzero singular values equals the rank, the column rank, and the row rank of a matrix.

10 Prompts to Explore with AI

6 Explain why truncated SVD is an approximation, and how this approximation error arises.

7 Give an example of how truncated singular value decomposition is applied in practice for image compression.

8 Explain why matrix V always stays square in the economy SVD of a tall matrix.

9 Explain the relationship between the eigenvalues of $A^T A$ and the singular values of A .

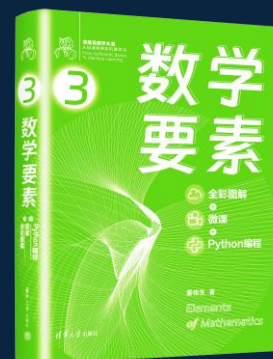
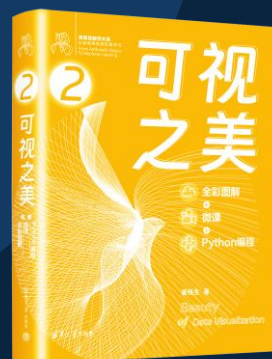
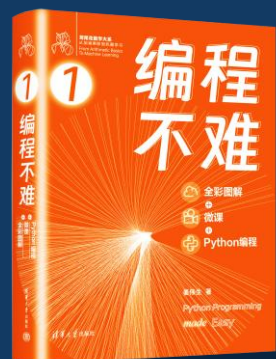
10 Compare the storage efficiency of full SVD and compact SVD, and explain how to choose the right SVD form.

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